

GETTING STARTED

UNPACKING

All SignalCore products ship in antistatic packaging (bags) to prevent damage from electrostatic discharge (ESD). Under certain conditions, an ESD event can instantly and permanently damage several of the components found in SignalCore products. Therefore, to avoid damage when handling any SignalCore hardware, you must take the following precautions:



- Ground yourself using a grounding strap or by touching a grounded metal object.
- Touch the antistatic bag to a grounded metal object before removing the hardware from its packaging.
- Never touch exposed connector pins.
- When not in use, store all SignalCore products in their original antistatic bags.

Remove the product from its packaging and inspect it for loose components or any signs of damage. Notify SignalCore immediately if the product appears damaged in any way.

Verifying the Contents of your Shipment

In addition to this Getting Started Guide, verify that your SC5506A kit contains the following:

Quantity	Item
1	SC5506A Dual Channel RF Signal Source
1	USB Flash Drive Installation Software (may contain multiple, individual product folders)

SYSTEM REQUIREMENTS

Supported Operating Systems	• Windows 7 Professional & Ultimate (both 32-bit and 64-bit versions) • Windows Vista (all editions) • Windows XP Professional (32-bit version only) • Windows 2000 Service Pack 3 or later • LabVIEW RT ETS (x86) The SC5506A device driver <u>is not supported</u> on the following operating system versions: • Windows NT/CE/Mobile/Me/98/95 • Windows XP x64 edition
RAM	256 MB minimum (required by LabVIEW)
USB Port	Installing device driver software, device applications, and example code

SOFTWARE INSTALLATION

SignalCore products are best installed into a system using a “software-first” installation order. Careful attention to the documentation installation order will ensure a successful device installation for your system. **Do not attach the SC5506A hardware to the host computer at this time.**



LabVIEW™ 2009 or greater and/or the LabVIEW™ Run-time Engine are required to view and run any LabVIEW™ examples, install and use the LabVIEW™ palette function libraries provided by SignalCore, or to run the SC5313A Soft Front Panel application.

1. Run "SC5506A Installer.exe" located under the SC5506A directory on the supplied USB flash drive.
2. If LabVIEW or the LabVIEW Run-Time Engine are not installed on your computer, select "Install LabVIEW Run-Time Engine" on the installation menu. This file is required for the Soft Front Panel application to run properly.
3. After the LabVIEW installer finishes, exit the LabVIEW installation window to return to the main SC5506A installation menu. The LabVIEW Run-Time installer may require you to reboot your computer to complete its installation. If so, once the computer has booted and returned to the desktop, navigate back to the SC5506A folder on the USB flash drive and proceed to the next step.
4. On the main installation menu, select either "Install SC5506A Software for 32-bit Windows" or "Install SC5506A Software for 64-bit Windows" depending on which type of operating system is installed on the host computer. Follow the instructions in the installer application. Exit the main SC5506A installation menu when finished.

SC5506A Device Driver Installation

To install the necessary device driver, please follow the instructions beginning with step 6 in the *SC5506A_ReadMe.txt* file located under the *SC5506A\Win* directory on the USB installation flash drive.



The SC5506A device drivers are not digitally signed. However, SignalCore takes proper precautions and all device drivers are thoroughly tested. If any security alerts pop-up during installation, please select the "Install this driver software anyway" option.

For LabVIEW support, an additional LabVIEW API is also provided and is available in the *Win\API\LabVIEW* directory. To copy the function libraries to the host computer, please follow the instructions beginning with step 19 in the *SC5506A_ReadMe.txt* file located under the *SC5506A\Win* directory on the USB installation flash drive.

HARDWARE INSTALLATION

1. Install the necessary device driver first, following the instructions detailed in the previous section.
2. Apply power to the device. The device requires a nominal supply voltage of +12 VDC, however it will handle supply voltages up to +15 VDC and as low as +10 VDC. It is important the power supply is capable of providing nominal power of 20W. The average device power consumption is 16 W but could consume as much as 18W initially. Ensure the supply polarity is correct prior to turning on the power supply. The device has some level of reverse power protection, however it is always good practice to check polarity first before applying power.
3. Connect a mini-USB to Type A USB cable between the device and an available USB port on the host computer.
4. Verify that the front panel "status" LEDs are green. Your SC5506A is now ready to use. If either of the "status" LEDs are not green, contact SignalCore technical support immediately.
5. Provide sufficient cooling for the device at all times. With proper cooling, the internal module temperature, displayed on the Soft Front Panel application or accessible programmatically through the API, should read approximately 15 °C above ambient and should never be allowed to exceed a maximum of 70 °C.